

CHAPTER 6

Intercellular Communication

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Orchestrating the activity of cell populations for physiological functioning of the brain, organs, and musculature depends on transmission of signals, learning and memory devices, and feedback control systems. By what biophysical mechanisms do cells communicate in order to coordinate their activity as local ensembles, as multimodal circuits, and across system levels? Here we only scratch the surface of this fascinating topic. We will focus on electrically active cells; for this, you can have in mind, for example, cardiac cells, many types of secretory cells, and neurons.

We know that ionic currents underlie cellular electrical activity. Hence, one way that cells can interact is by directly passing ionic current between each other. Perhaps the simplest mechanism for such communication is the analogue of resistive coupling between units, i.e., with the intercellular current being proportional to the voltage difference between cells. Heart cells, and many other types of cells, communicate in this manner, with the ions flowing directly between two coupled cells. In this case, referred to as electrical coupling (and sometimes, in the neural context, electrotonic coupling), the current flows through channel proteins that span the plasma membranes of both cells as shown in Figure 6.1. The clusters of such channel proteins that are found at cell-to-cell contacts are called gap junctions.

While gap junctions are occasionally found in neural circuits, they seem to be less uncommon during development or they may not constitute an exclusive means of interaction. Neurons have a rich repertoire of other ways for exciting or inhibiting other target neurons, via the indirect means of chemical synaptic transmission depicted in Figure 6.5. At a terminal of a neuron's axon the neurotransmitter of one type or another is bundled in vesicles that are released with increased probability when the terminal is

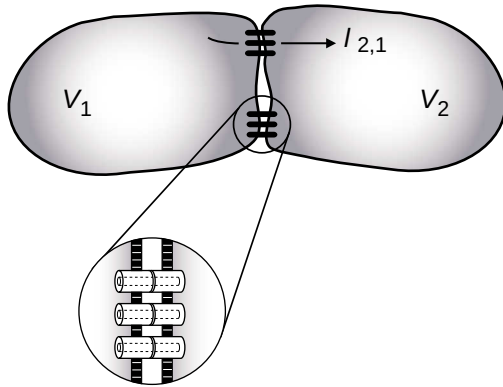


Figure 6.1 Schematic of gap junctions between two cells. Coupling current from cell #1 to #2 is $I_{2,1} = g_c(V_2 - V_1)$, the exact negative of $I_{1,2}$. Redrawn from [Hille, 2001].

active (depolarized). The transmitter, released into the extracellular space in the terminal region, diffuses and binds to receptors on the postsynaptic cell's nearby membrane. These receptors may be part of a receptor-channel complex or be linked indirectly through second messengers to a nearby channel. The activated channels then lead to the postsynaptic current and action. The variety of transmitters and receptors enable many different time scales and "sign" of the input that is being delivered to the target cell. We find synapses that are excitatory or inhibitory; they can be fast or slow; they can be shunting; their synaptic parameters can change with usage, either, depressing or facilitating; they can be voltage gated or not on the postsynaptic side.

Gap-junctional coupling is typically localized, certainly for cells without spatially extended processes, to nearby neighbors. Prime examples include the heart and islets of Langerhans in the pancreas. In contrast, neurons can interact across distances that are many times greater than a cell body diameter, by means of their potentially far-reaching axonal and dendritic arbors. The synaptic interactions enable them to participate in local calculations and with distant assemblies, on selective time scales, fast or slow and to various degrees. For long-distance communication between cells propagated action potentials typically mediate signal transfer to the synapses via axons.

Given the variety of coupling mechanisms and connectivity patterns there are many possible behavioral modes, spontaneous and/or stimulus-driven, that such circuits may exhibit. Obviously we cannot elaborate on many of these behaviors; we will restrict attention to a few examples and focus on pairwise interactions. Also we choose not to present here some other mechanisms for intercellular interaction such as coupling by cell-generated electrical fields, diffusion and exchange of ions (that effect Nernst equilibrium potentials) and second messengers, mechanical and hormonal effects [Keener and Sneyd, 1998].

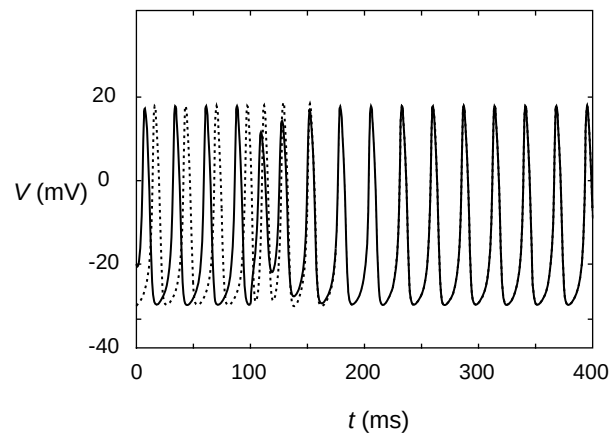


Figure 6.2 Two identical Morris-Lecar neuron models, auto-rhythmic, synchronize with electrical coupling (turned on at $t = 100$ ms). Parameter values for the model are those of Chapter 2 with the following exceptions: $v_3 = -5$ mV, $v_4 = 10$ mV, $\phi = 0.5$, $g_{Ca} = 8$ mS/cm², $I_{app} = -10\mu\text{A}/\text{cm}^2$. Net gap junction coupling conductance is $g_c = 2$ mS/cm². Initial conditions: $V_1 = -20$ mV, $w_1 = 0.2$, $V_2 = -30$ mV, $w_2 = 0$.

6.1 Electrical Coupling and Gap Junctions

The proteins that constitute the channels at gap junctions are of the connexin family. Connexin molecules in the plasma membrane of one cell link up those in an adjacent cell to form the channels. These channels pass most ions as well as various molecules up to molecular weights of tens of kilodaltons, including those involved in second messenger systems such as IP₃. The single channel conductance can be in the range of 75–150 pS, and in most known instances they are relatively voltage-independent. The variety of connexins provides a rich repertoire for diverse modulatory possibilities by changes in intracellular pH, second messengers, neurotransmitters, or voltage [Perez Velasquez and Carlen, 2000]. For example, the reduction of gap-junctional coupling due to increasing levels of Ca²⁺ is described as a protection mechanism.

For cell-level modeling we typically represent the net conductance of a gap junction as a constant, the product of the mean number of open channels at the junction and the single channel conductance. Simultaneous pairwise recordings, at least for geometrically simple neighboring cells, can be used to estimate this conductance. For spatially extended cells, like neurons with branching dendrites, such measurements would be confounded especially if the gap junctions are at electrically remote sites. Gap junctions may also be detected by dye coupling, although this method is problematic without good controls to test how readily the dye passes.

6.1.1 Synchronization of Two Oscillators

Consider the case of two cells idealized as isopotential compartments as shown in Figure 6.1. The gap-junctional current $I_{2,1}$ that flows from cell 1 into cell 2 is written as $g_c(V_1 - V_2)$, where g_c is the net coupling conductance of all the gap junctions formed between the two cells. This current appears as a source term in the current balance

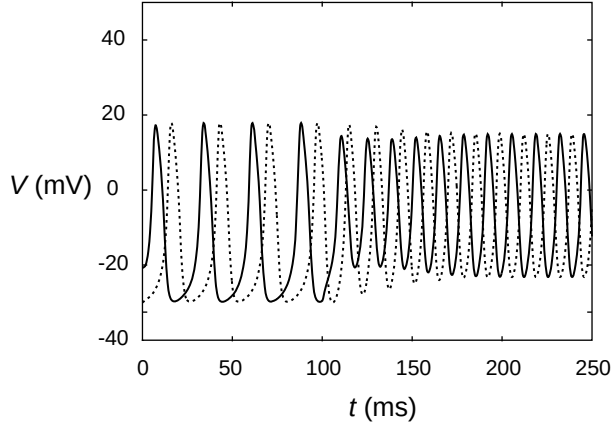


Figure 6.3 Time courses of V_1 , V_2 showing antiphase behavior when weak gap-junctional coupling is turned on at $t = 100$ ms. Same initial conditions and parameters as in Figure 6.2, except that here $g_c = 1$ mS/cm².

equation for cell 2. Defined as such, with the current as leaving from cell 1, it appears as a sink term for cell 1 (or equivalently, $g_c(V_2 - V_1)$ is a source term into cell 1). Thus we have

$$\frac{dV_1}{dt} = \frac{-I_{\text{ion}}(V_1, w_1) + I_{\text{app}} + g_c(V_2 - V_1)}{C_m}, \quad (6.1)$$

$$\frac{dV_2}{dt} = \frac{-I_{\text{ion}}(V_2, w_2) + I_{\text{app}} + g_c(V_1 - V_2)}{C_m}, \quad (6.2)$$

where w_1, w_2 correspond to the set of gating variables in each cell. Let us first predict the behavior, say in the case where the gap junction conductance is large. This means that the cells are very tightly electrically coupled. We would then expect them to have approximately the same voltage. Indeed if the cells are identical, then $V_1 = V_2$ is always a solution: The coupling current would be zero in this case. Of course, a perturbation (e.g., brief current pulse) to just one cell would make their V 's differ, transiently. By subtracting the two equations and then dividing by g_c we see, after dropping the term of order $1/g_c$, that

$$\tau_c \frac{d(V_1 - V_2)}{dt} \approx -(V_1 - V_2), \quad (6.3)$$

where $\tau_c = C_m/g_c$. This shows that the cells will re-establish uniformity with an effective time constant of τ_c , that is very short when g_c is large.

This is illustrated in Figure 6.2 for two Morris-Lecar cells that are tuned into an oscillatory regime (as discussed in Chapter 2). In this example the cells are initially out of phase and uncoupled. Synchronization occurs promptly after the coupling is introduced at $t = 100$ ms.

6.1.2 Asynchrony Between Oscillators

While electrical coupling is typically considered as a mechanism for uniformizing cells, one should be aware that the outcome depends considerably on coupling strength and

dynamic features of the individual cells. For example, the same cells used above when coupled weakly with gap junctions do not synchronize. Instead, they establish a rhythm with the two cells in a stable antiphase-locked pattern: just the opposite of togetherness (Figure 6.3). Also, for this situation the period of the network oscillation depends on the value of g_c . This is not true for the in-phase pattern. When the coupling strength is large enough, the period (for a pair of identical cells) is that of the isolated cell and is therefore independent of g_c .

6.1.3 Cell Ensembles, Electrical Coupling Length Scale

Such synchronization also occurs for spatially distributed multicell systems; even though electrical coupling is localized it can, when large enough, synchronize a population that might be spread over a sizable spatial region. In order to formalize this, one can introduce the concept of an electrical length scale that involves coupling conductance as well as leakage conductance. This emphasizes the relativeness of intercellular current to the current flowing across a cell's plasma membrane. By using this notion we can refer to the “electrical size” of a cellular array; it could be quite compact even though the spatial extent might not be.

In order to formalize this notion we consider the simple case of Figure 6.4, where we have a chain of gap-junction-coupled cells that have passive membrane properties, i.e., $I_{\text{ion}} = g_m V$. Thus we have the equation for a typical cell, cell j :

$$\frac{dV_j}{dt} = \frac{g_c(V_{j-1} - V_j) - g_m V_j + g_c(V_{j+1} - V_j)}{C_m}. \quad (6.4)$$

Suppose that we have a very long chain of cells (and for now, ignore end effects) and suppose we voltage clamp the cell in the middle for which $j = 0$ ($j > 0$ refers to cells to the right and $j < 0$ is for cells to the left). In this passive system the voltage distribution will go to a steady state after some transient, and the voltage will decrease from V_0 as $|j|$ increases. This attenuation occurs as current spreads from one cell to the next away from $j = 0$ and as current is lost through the “leaky” membrane of each cell.

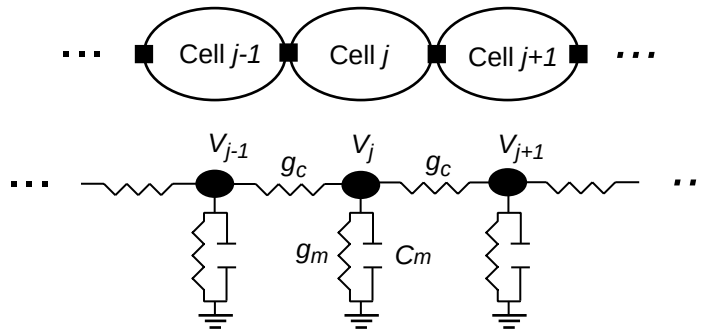


Figure 6.4 Schematic of linear array of passive cells coupled (nearest neighbor) by gap junctions.

The steady voltage decays in an exponential fashion with distance. The decay “rate” γ can be found by considering the steady-state case of (6.4) (i.e., setting $dV_j/dt = 0$) and seeking a solution of the form $V_j = V_0\gamma^j$. This leads to a quadratic equation for γ :

$$\gamma^2 - (2 + g_m/g_c)\gamma + 1 = 0. \quad (6.5)$$

Hence, there are two values of γ :

$$\gamma_{\pm} = 1 + g_m/(2g_c) \pm \sqrt{[1 + g_m/(2g_c)]^2 - 1}. \quad (6.6)$$

Note that both roots are positive with $\gamma_+ > 1$ and $\gamma_- < 1$.

For a voltage profile that decays with distance to the right of $j = 0$ we must disallow γ_+ (such a term would grow with j). Thus, if we ignore end effects due to termination of the chain, the steady state profile for $j \geq 0$ has the form

$$V_j = V_0\gamma_-^j. \quad (6.7)$$

The solution for $j \leq 0$ is $V_j = V_0\gamma_+^j = V_0\gamma_-^{-j}$, since $\gamma_+\gamma_-^{-1} = 1$.

We can test our intuition by considering some limiting cases. When g_m/g_c is large, the leakage conductance is dominant and the voltage should attenuate rapidly with distance from $j = 0$. Indeed, $\gamma_- \approx g_c/g_m$, so that V_j decays steeply. On the other hand, when g_m/g_c is small, then $\gamma_- \approx 1 - \sqrt{g_m/g_c}$ and the spatial decay is gradual.

Pursuing this latter case a bit more, if cell-to-cell attenuation is small and the spatial profile changes smoothly, we might treat the one-dimensional cellular array approximately as a continuum with position $x \approx j\Delta x$, where Δx is a cell-size parameter, considered small. The leakage conductance g_m is proportional to a cell’s surface area, which is the perimeter of a cross section times Δx . We will take some care with the representation of g_c . Let’s think about the reciprocal, $r_c = 1/g_c$, the resistance to longitudinal current flow between cells, say from one cell center to the next. Then, r_c involves resistance due to the gap junctions as well as cytoplasmic resistance along the cell’s axial direction. If there are many gap junctions between adjacent cells, then the major contribution to r_c will be the cytoplasmic term, which is proportional to the area of a cross section times Δx . Consequently, the ratio $g_m/g_c (= g_m r_c)$ is proportional to Δx^2 and may be written as $(\Delta x/\lambda)^2$. Finally, combining this with our expression for γ_- (from (6.6) for small g_m/g_c) and noting that $\log(\gamma_-) \approx -\sqrt{g_m/g_c}$, we see that the solution form for the continuum approximation is

$$V(x) = V_j = V_0 \exp(j \log \gamma_-) \approx V_0 \exp(-j\Delta x/\lambda) = V_0 \exp(-x/\lambda). \quad (6.8)$$

Thus, λ is the electrical length constant for the continuum approximation.

This treatment relates nicely to continuum models for current spread in individual cells that have extended processes, like neurons with their dendrites or axons. These processes behave like electrical cables; current flows along the axis and leaks through the membrane. In this case of a neuronal process, we could think of it for simulation purposes as a chain of short compartments (length Δx) each with a membrane surface area and connected by axial resistances; there are no gap junctions between the

fictitious adjacent compartments. If d is the diameter of the cross section, then we would have $g_m = \Delta x \pi d G_m$, where G_m is the leak conductance density (S/cm²) and $r_c = \Delta x 4 R_i / (\pi d^2)$ where R_i is the cytoplasmic-specific resistivity (ohm·cm). Putting this together we obtain for the electrical length constant of a passive neuronal dendrite or axon $\lambda = \sqrt{d / (4 G_m R_i)}$. As an illustration, for some neurons estimates are in the range $G_m = 0.25$ mS/cm², $R_i = 100$ Ω·cm, so that with $d = 2.5$ μm yields $\lambda = 500$ μm. This means that for a dendritic branch of physical length 250 μm, the corresponding electrotonic length is about 0.5. Therefore, we do not expect excessive attenuation for steady or slow voltage changes. Transients decay more abruptly. Also, the attenuations are more severe from dendritic locations to the cell body than vice versa. In electrically coupled excitable systems with weaker gap junctions or extended electrical size (physical length larger than λ) synchrony is not quickly established, and localized perturbations may lead to waves or other spatiotemporal patterning.

The treatment of dendrites as cables and the compartmental method is discussed in Chapter 7, and in more depth in [Johnston and Wu, 1995], [Koch and Segev, 1998], [Koch, 1999], and [Keener and Sneyd, 1998]. We note that this discretization of neuronal cable-like processes is called the compartmental method and was first developed by Rall for treating the effect of spatiotemporally distributed synaptic inputs over a neuron's somatic-dendritic area [Segev et al., 1995].

6.2 Synaptic Transmission Between Neurons

In the preceding section we saw that strong gap junctions can be used for coordinating cellular electrical activity, subserving functions in which cells might work together in approximate synchrony. The bidirectionality and instantaneous nature of gap-junction coupling are well suited for achieving these goals. In the nervous system there are demands for more complex patterning. Individual cells generally are likely involved in numerous different computations, with possibly different time scales, sometimes being called into action with a brief wake-up call but in other cases only after a long barrage of inputs. In some cases the precise timing of action potentials might be important, and we would expect fast coupling mechanisms. But when firing rate, rather than spike timing, is more important for signaling, perhaps the synchronization properties of fast coupling may be less critical and slower coupling mechanisms might dominate. Some flexibility for multiplexing of these signaling modes is attainable by exploiting the potentially large variety of intrinsic mechanisms, i.e., the many different ionic channels and modulators for them. Even greater computational power can be achieved, however, by selective adjustment of the many synapses on a cell, maybe in an activity-dependent fashion. One can imagine a great many neural ensembles with possibly shared units that can be dynamically constructed, recruited, and dismissed as needed for memory storage, sensory processing and perception, and executive command functions.

It is hard to imagine how instantaneous bidirectional coupling could allow in an efficient way such a rich set of alternatives. However, the one-way signaling via chemical

synapses enables the system to employ a few transmitters but yet enrich the possibilities for postsynaptic response by having many choices for the postsynaptic receptors and channels. Sites for modulatory action can be implemented on either the postsynaptic or presynaptic side. Thus while vesicle-packaged transmitter is released in punctate fashion, like the action potential, the time scale and even the “sign” of the response are determined by the machinery on the postsynaptic cell.

6.2.1 Kinetics of Postsynaptic Current

We will formulate an idealized model for the current generated in a postsynaptic cell due to transmitter release by a presynaptic cell in which an action potential has occurred. The sender's action potential opens voltage gated calcium channels at the axon terminal where vesicles are poised ready to fuse with the membrane and release transmitter.

Whether or not release occurs is a probabilistic event. The failure rate can be high at some, e.g., cortical, synapses. There is also a small probability of spontaneous release. Experiments at the neuromuscular junction and central synapses use the quantal release hypothesis to analyze the statistics of postsynaptic responses for spontaneous and evoked release to estimate the number of active sites, quantal content, and release probability [Johnston and Wu, 1995]. We will not consider these issues here, nor the details of transmitter diffusion in and removal from the synaptic cleft, the small extracellular space between the pre- and postsynaptic sites.

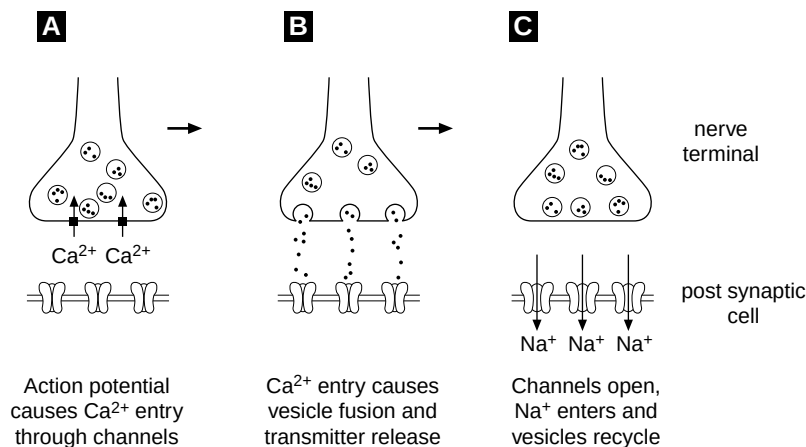
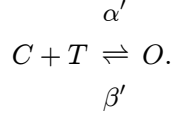


Figure 6.5 Steps in chemical synaptic transmission. Action potential arrives at terminal of presynaptic axon; depolarization opens voltage gated calcium channels. Resulting calcium influx causes vesicle fusion and transmitter release. Transmitter diffuses, reaches postsynaptic membrane, and binds to receptors. The activated receptors cause postsynaptic channels to open. Depending on which type of ions are allowed to pass (sodium shown here) the postsynaptic response can be depolarization or hyperpolarization. Adapted from [Koch, 1999].

We suppose that transmitter is available briefly, and that during this time it can bind to receptors, actually receptor–channel complexes, on the postsynaptic membrane. For this simple model we imagine a two-state channel. Binding of transmitter to a receptor molecule favors opening of the channel and unbinding leads to closing. A kinetic scheme for this is the following:



If s is the fraction of complexes in the open state O , then $1 - s$ is the fraction closed, and we have, given a transmitter concentration $[T]$,

$$\frac{ds}{dt} = \alpha'[T](1 - s) - \beta's. \quad (6.9)$$

This representation assumes unlimited transmitter and receptor availability. Also, it requires us to specify a time course for $[T]$; for example, we might assume a square pulse over some time duration (perhaps with a modest delay, on the order of a millisecond or less) when the presynaptic cell fires. We will adopt a simpler, and easy to implement, scheme, supposing that maximal transmitter is available when the presynaptic membrane potential V_{pre} is above some level, say θ_{syn} . This would mean $[T] = T_{\text{max}}s_{\infty}(V_{\text{pre}})$ with $s_{\infty}(V_{\text{pre}}) = H(V_{\text{pre}} - \theta_{\text{syn}})$, where $H(x)$ is the Heaviside step function. We will smooth this out using a sigmoidal function for s_{∞} . Redefining the rate constants, which we view as adjustable depending on the desired time course of postsynaptic conductance, we get

$$\frac{ds}{dt} = \alpha s_{\infty}(V_{\text{pre}})(1 - s) - \beta s, \quad (6.10)$$

with $s_{\infty}(V) = 1/[1 + \exp(-(V - \theta_{\text{syn}})/k_{\text{syn}})]$, with k_{syn} positive. The current through the post-synaptic membrane is

$$I_{\text{syn}} = g_{\text{syn}}s(V - V_{\text{syn}}), \quad (6.11)$$

where the reversal potential V_{syn} depends on the concentration differences for the ion species that flow through the open receptor channels in the postsynaptic membrane; g_{syn} is the maximal conductance at the synapse if all the available channels are open.

6.2.2 Synapses: Excitatory and Inhibitory; Fast and Slow

This type of model has been used to describe some common types of excitatory and inhibitory synapses that have relatively simple kinetics. The different transmitters and different receptors and different ions that pass through the synaptic channels lead to a variety of synaptic types. The most commonly known transmitter that is used for excitatory synapses is glutamate. It can activate AMPA receptor channels that typically pass inward current (carried mostly by sodium ions, but also some potassium and other

ions). The postsynaptic conductance is relatively fast with a rise time ($\approx \alpha^{-1}$) of order 1 millisecond and decay times ($\approx \beta^{-1}$) of a few to tens of milliseconds. Expressions such as (6.10) have been used for these synapses. The value of V_{syn} in this case is typically about 100 mV above rest. Glutamate can also evoke slower postsynaptic responses by activating NMDA receptors (fast rise, of order milliseconds, and slow decay, of order many tens to hundreds of ms) or a number of different types of metabotropic glutamate receptors with very slow rise and slow decay of order seconds. The conductance associated with NMDA receptors is, curiously, also gated by the postsynaptic voltage V_{post} . If V_{post} is not adequately above V_{rest} , the NMDA-associated channels are blocked by magnesium ions from the outside; sufficient depolarization relieves this block, and the channels can open. The dependence of synaptic current on pre- and postsynaptic activity implicate the NMDA conductance in various models for associative learning and synaptic plasticity generally. The very slow metabotropic glutamate-mediated effects are believed to be primarily modulatory, acting, for example, to effectively change the firing threshold of a cell. In the examples below of excitatory synaptic coupling we will be thinking primarily of AMPA-mediated excitatory synapses, for which (6.10) is a good first approximation. More complete descriptions, even for this simple receptor-channel complex, might involve kinetic models with additional states [Destexhe et al., 1998].

Fast inhibitory synapses are activated by the transmitter GABA when they bind to GABA_A receptor channels. The postsynaptic conductance has a fast rise time, one or a few milliseconds, and somewhat slower decay than the AMPA-excitatory conductance, on the order of tens of milliseconds. The current is carried largely by chloride ions. Interestingly, the value for V_{syn} can be quite variable, typically -60 mV to -80 mV but in some cases as depolarized as -35 mV; this variability likely reflects different types or activities of chloride pumps that regulate intracellular Cl^- concentration. If a cell is sitting near its resting potential of say -65 mV, the synaptic current that is generated by activating these channels would be outward (hyperpolarizing) if V_{syn} is below -65 mV or inward (depolarizing) if V_{syn} is above -65 mV. It is not uncommon for GABA_A -mediated currents to be depolarizing in developing neural tissue. Thus it is not strictly correct, although frequently done, to refer to GABA as an inhibitory transmitter. One should take care in describing a GABA_A -mediated synapse as inhibitory; the current's sign depends on where V_{syn} is relative to V . Note that if V is close to V_{syn} , little synaptic current will be generated, even if a large conductance g_{syn} is activated. On the other hand, the membrane potential would be effectively clamped to V_{syn} until the conductance deactivates. In this case the GABA_A synapse acts as a strong shunt in the membrane; other modest-sized synaptic inputs would be ignored during this time.

The transmitter GABA acting through GABA_B receptors can lead to a very slow inhibition with a conductance that rises and decays on the order of 100 or more milliseconds. The current is carried primarily by K^+ ions and so the reversal potential may be -70 to -90 mV. The simple model in (6.10) cannot account for this slow current. One shortcoming is that the model predicts a peak shortly after the depolarization of V_{pre} , while GABA_B -mediated inhibition peaks only much later. A minimal model for this current would involve at least two dynamic variables.

Typical sizes of I_{syn} at a single synapse are a few picoamperes and may evoke responses (e.g., in cortical neurons) of 0.1 to 1 mV. A number of inputs must be summed in order to bring the neuron to firing threshold. Generally, the postsynaptic response to multiple inputs, even in the subthreshold regime, is not, however, a linear summation of individual inputs. This is because the synaptic current depends on V_{post} , in particular through the driving force $V_{\text{post}} - V_{\text{syn}}$. If V_{post} is far from V_{syn} , then the dependence on V_{post} is very weak, as, say, for AMPA-mediated excitation if V_{post} is near rest. But in other cases, say for a GABA_A-mediated synapse, we would not get linear summation: Doubling, say, the number of inputs does not double the synaptic current. This is easy to see for a passive membrane system, as we considered above, with steady synaptic conductance input $g_s = \text{constant} = g_{\text{syn}} s$. The steady response $\bar{V}$ would be

$$\bar{V} = V_{\text{syn}} g_s / (g_m + g_s), \quad (6.12)$$

saturating for large g_s at V_{syn} , as we expect. While increasing g_s drives V_{post} closer to V_{syn} , the increment in synaptic current diminishes for large g_s : dI_{syn}/dg_s behaves like g_s^{-2} for large g_s ; this is the sublinear summation effect.

When synaptic inputs are activated repeatedly the response may not remain uniform from one stimulus to the next in a train. This dynamic aspect may involve facilitation or depression of successive responses on a range of time scales. For example, the response might be depressed by 50% or more to a steady level after less than ten stimulus repeats (say at 50 Hz), and then recovery may take a second or so [O'Donovan et al., 1998]. A favored mechanism for depression is presynaptic depletion of ready-to-release transmitter. Synaptic depression is analogous to receptor desensitization in other contexts, such as hormonal communication [Goldbeter, 1996].

Various possible functional roles for such synaptic dynamics in network activity have been suggested. One hypothesis is that synaptic depression could mediate phasic behavior of a feedforward network. That is, if the steady response saturates beyond some input rate (so the tonic output conveys no information about the input rate), the network could still be responsive to transient changes in input rate; just after a sudden increase in input rate the synapses would not yet be adapted to their new target level of depression, so the network's output would briefly rise and then fall back to the saturation level. In the context of recurrently connected networks synaptic depression provides a mechanism of slow negative feedback for the intercellular coupling. This can underlie rhythmic population activity even if isolated units are capable only of tonic firing for steady inputs; here the autocatalysis could be provided by recurrent excitatory connections [Tabak et al., 2000].

6.3 When Synapses Might (or Might Not) Synchronize Active Cells

In order to illustrate some effects of mutual synaptic coupling between cells we will consider the simple case of two identical cells, each of which is autorhythmic, and

examine conditions for which the cells tend to fire together or apart. We are asking about the synchronization patterns in this simple two-cell network. Classical notions are that mutual excitation tends to make cells synchronous (in phase), while mutual inhibition pushes them apart, leading to antiphase behavior. That is, if two excitatory cells are somewhat out of phase, the firing of the leading cell encourages the follower to fire sooner, thus bringing the cells more nearly in phase, while for inhibitory cells that are somewhat out of phase the leading cell's firing will delay the follower, increasing their phase difference. Successive cycles would increase their phase difference to 180° but not beyond, since then the follower would act as the leader, pushing the phase difference back toward 180° . These expectations and some surprises will be illustrated below when we consider the effects of synaptic time scales.

Here we will use the Morris-Lecar model in Type I mode (see Chapter 2), so that near the threshold for repetitive firing the steady firing frequency can be made arbitrarily low. We dictate here that the gating variable's kinetics are relatively slow and that the stimulating current is adjusted so that the cell model is firing slowly, about 15 Hz.

First we consider the case of mutual excitation. In panels A and B of Figure 6.6A we confirm the expectation that the cells will fire together. The cells are substantially out of phase before we actually implement the coupling, but then afterwards they converge to a pattern of near synchrony, with one cell preceding the other by just a slight bit. When the coupling is first turned on the follower cell immediately advances, and throughout the transient phase of synchronization the cells are causing each other to fire faster. Interestingly, as synchrony is established the cells slow down to nearly their intrinsic frequencies, as if the coupling is only to synchronize them. When comparing the time courses we see that during the transient phase the leader cell's voltage is more affected than the follower's. The synaptic current from the follower at first reduces the leader's postspike hyperpolarization. Then, as the follower catches up, this synaptic current is delivered when the leader is still strongly depolarized, and its intrinsic conductances swamp the perturbing effect of the synaptic input.

For the preceding example the synaptic conductance time course has fast rise and fast decay phases; the conductance is essentially activated only during the presynaptic depolarization. If we allow for the synaptic decay to be much slower, with all other parameters and initial conditions unchanged, we find that these two cells now fire in antiphase, as shown in Figure 6.6C and D. The frequency of each cell is nearly twice that of an isolated cell. This is understandable. Since the synaptic current decays more slowly, it provides a longer-lived depolarizing influence, and it is strongest during a cell's trajectory as it rises toward threshold, when it is most responsive to depolarizing influences. Note that if the output of this two-cell network converges onto a common target, the effective delivery rate, because of the antiphase pattern, is twice that of each, four times that of an individual cell.

For the examples in this section we have chosen parameter values in order to emphasize the importance of synaptic time scales in determining firing patterns in networks. This is an active area of research; for example, see [Wang and Rinzel, 1992,

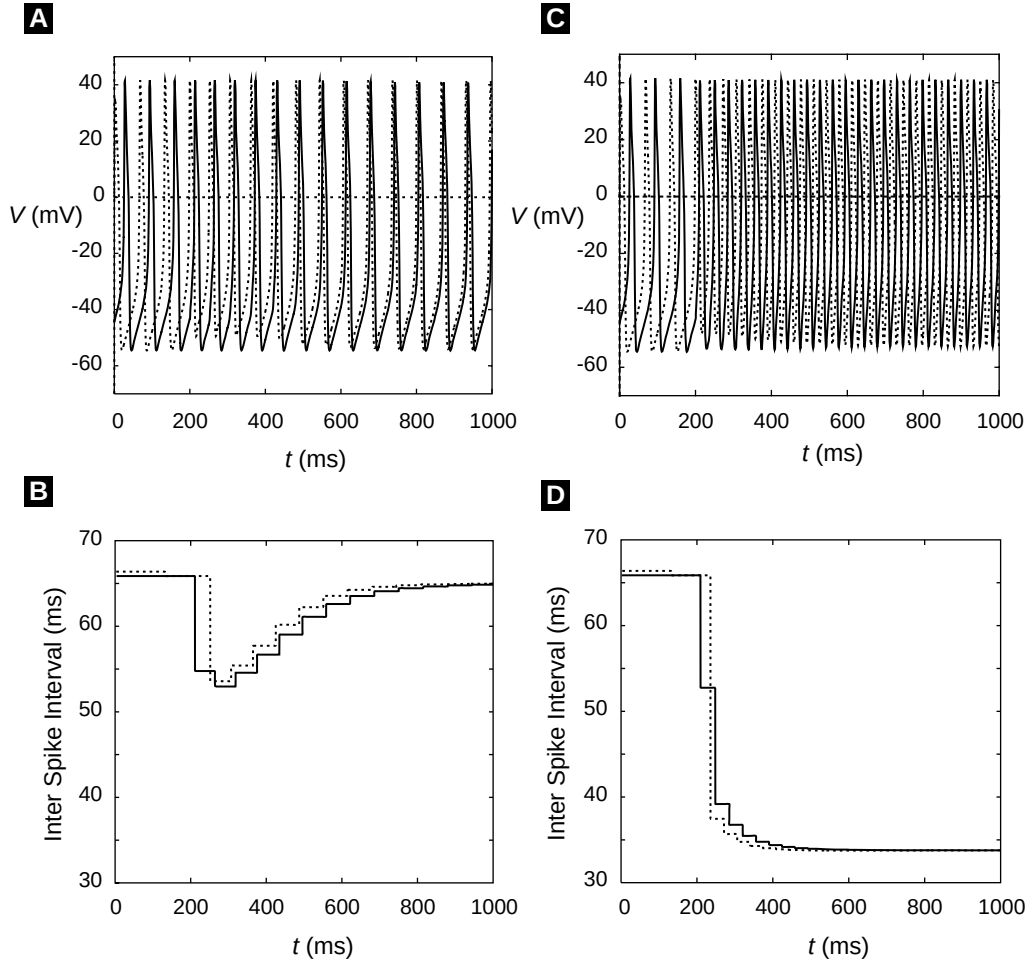


Figure 6.6 The kinetics of synaptic excitation affect synchronization properties of cells, as illustrated with two identical coupled cells. Neuron models are uncoupled until $t = 200$ ms. (A and B) fast excitatory synapses lead to synchrony; time courses of V_1, V_2 in (A) and of instantaneous interspike interval in (B). (C and D) antiphase pattern evolves with slow excitatory synapses; time courses of V_1, V_2 in (C) and of instantaneous interspike interval in (D). Parameter values for v_3, v_4, g_{Ca} are as in Figure 6.2–Figure 6.3, except that here, $\phi = 0.1$ and $I_{app} = 15 \mu A/cm^2$; for the synaptic kinetics, $\theta_{syn} = 20$ mV, $k_{syn} = 2$ mV with $\alpha = 3/ms$, $\beta = 1/ms$ in (A and B) and $\alpha = 3/ms$, $\beta = 0.1/ms$ in (C and D). Initial conditions: $V_1 = -44$ mV, $w_1 = s_1 = 0$, $V_2 = 0$ mV, $w_2 = 0.3$, $s_2 = 0.3$.

Vreeswijk et al., 1994]. Deeper understanding will be achieved, and for now, we caution that it should not be taken as universal that fast (slow) excitatory synaptic coupling leads to in-phase (antiphase) firing. In fact, for the example in Figure 6.6A and B, different initial conditions to such coupled cells can lead to antiphase locking even for these fast synapses.

Next we consider the case of two cells (the same cells as above) coupled with inhibitory synapses. We start the cells with different initial voltages and watch them settle slowly into an antiphase rhythm (Figure 6.7) during the first 800 ms of the simulation. During this portion the inhibitory synapses have fast kinetics, and we are confirming the classical expectation that inhibition leads to antiphase firing between a cell pair. In these voltage time courses one can clearly see the hyperpolarizing effect of the synaptic inputs. As a cell's voltage rises from its minimum the partner fires and sends a brief pulse of outward current, which halts transiently the rising voltage. At $t = 800$ ms the decay rate of inhibition is slowed from $\beta = 1/\text{ms}$ to $\beta = 0.1/\text{ms}$. Within a few cycles the cells lock into perfect synchrony. Slowly decaying mutual inhibition can lead to in-phase locking among neural oscillators. This behavior has been proposed as the mechanism for gamma rhythms that are seen in various brain regions, and are believed to have a functional role in some cognitive processes [Wang and Buzsaki, 1996, Traub et al., 1996].

As a secondary note, although we do not show it here, this particular network with the fast inhibitory synapses is bistable. In addition to the antiphase behavior (for $t < 800$ ms) it also has a stable in-phase behavior for some set of initial states.

The examples above are highly idealized and primarily directed toward rhythmicity. The parameters have been chosen in order for us to illustrate several features about temporal patterning in mutually coupled pairs with just minimal adjustments from one case to the next. Important questions arise about how these features might carry over to larger networks. In a case of mixed fast and slow synaptic coupling are the synchronizing or desynchronizing effects more important? Are the patterns robust to effects of noise and heterogeneity? What if there is a mixture of gap-junctional and synaptic coupling, in regimes where their effects counteract each other?

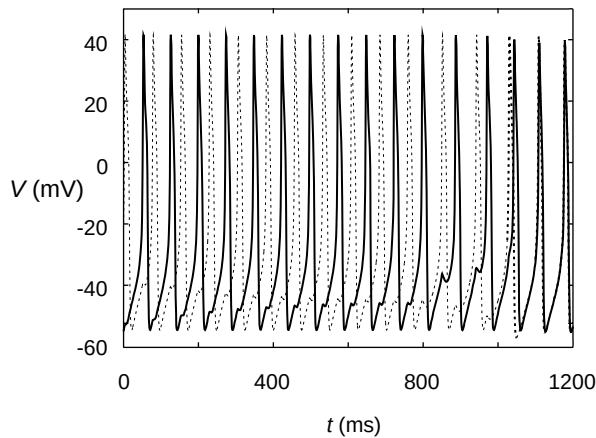


Figure 6.7 Kinetics of synaptic inhibition can affect synchronization. Fast inhibition leads to antiphase behavior for $t < 800$ ms; switching to slowly decaying inhibition for $t > 800$ ms leads to in-phase behavior. Parameters are the same as in Figure 6.6 except that here $\beta = 1/\text{ms}$ for $t < 800$ ms and $\beta = 0.1/\text{ms}$ for $t > 800$ ms. Initial conditions: $V_1 = -55$ mV, $w_1 = s_1 = 0$, $V_2 = -20$ mV, $w_2 = s_2 = 0$.

6.4 Neural Circuits as Computational Devices

Up to this point we have examined how various coupling mechanisms can coordinate cellular electrical activity in the context of stereotypical steady firing patterns: synchronous or antisynchronous oscillations. Beyond these examples one seeks to understand, in the context of neural systems, how circuits (cells and their connectivity) might be implemented to carry out various computations or computational strategies.

We will develop examples to illustrate idealized mechanisms for carrying out a few computational tasks: temporally specific addition or coincidence detection, temporally specific subtraction or vetoing, and direction selectivity. The setup will be similar in each case. There will be two input units, R_1 and R_2 , that we might think of as sensory neurons, receptors to the outside world. They feed into a target neuron that does the computation. (Figure 6.8A). This is a feedforward circuit. The information will be passed by means of chemical synapses. Depending on the combined synaptic input (triggered by the firings of R_1 and R_2) the target neuron may or may not elicit a spike. The “answer” from the computation will simply be yes or no, whether the target neuron fires or does not fire an action potential for our specified input pattern. This will depend on whether R_1 and R_2 are both activating excitatory synapses (Figure 6.8A) or one of them is sending inhibition and the other excitation (Figure 6.8B), and of course the relative timing will matter.

For a neuron in which inputs are distributed over the dendritic tree, the spatial as well the temporal distribution of incoming signals determines the net response. For example, suppose there is simultaneously timed excitation and inhibition delivered at different locations along one dendritic branch. An inhibitory synapse located closer to the soma will be more effective at reducing the response to the excitatory input than if it is located more distally on the branch.

Let $s_1(t)$ and $s_2(t)$ be the (fractional) postsynaptic conductances that arise in the target cell from activity of R_1 and R_2 , respectively. We will consider the results of only a single firing of each of the input neurons. The conductances $s_1(t)$ and $s_2(t)$ will be determined from the kinetics as in (6.10). The kinetic parameters are chosen so that an excitatory conductance time course is brief and an inhibitory conductance is slower, on time scales comparable to spike dynamics. For simplicity we do not describe the action potentials of R_1 and R_2 in detail. Instead, we represent the receptor voltage responses, $V_1^R(t)$ and $V_2^R(t)$, as brief (2 ms), square-pulse, depolarizing events. These will yield appropriate time courses for $s_1(t)$ and $s_2(t)$.

The parameters for our Morris–Lecar target neurons are the standard ones from Table 2.4 for Type II behavior; here, we set $I_{app} = 0$. The equations that govern the response of the target neuron then are as follows:

$$C \frac{dV}{dt} = -I_{ion}(V, w) - g_1 s_1(V - V_{syn,1}) - g_2 s_2(V - V_{syn,2}), \quad (6.13)$$

$$\frac{dw}{dt} = \frac{\phi[w_\infty(V) - w]}{\tau_w(V)}, \quad (6.14)$$

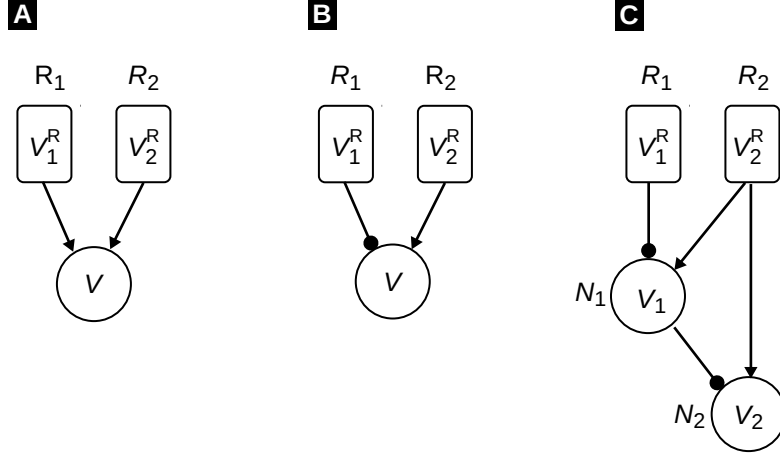


Figure 6.8 Schematic of receptors (R_1 , R_2), neurons and connections for implementing neural computations. (A) R_1 and R_2 activate excitatory inputs on target neuron. (B) R_1 sends inhibition; R_2 sends excitation. (C) First-layer neuron N_1 (same configuration as in (B)) sends inhibition to second-layer neuron N_2 , which also receives direct excitation from R_2 .

$$\frac{ds_1}{dt} = \alpha_1 s_\infty(V_1^R(t))(1 - s_1) - \beta_1 s_1, \quad (6.15)$$

$$\frac{ds_2}{dt} = \alpha_2 s_\infty(V_2^R(t))(1 - s_2) - \beta_2 s_2. \quad (6.16)$$

The parameters g_1 and g_2 represent the synaptic weights, the maximum postsynaptic conductances that can be evoked by activity coming from R_1 and R_2 . The values of $V_{\text{syn},1}$ and $V_{\text{syn},2}$, the synaptic reversal potentials, will determine whether the synapse is excitatory (with V_{syn} far above V_{rest} , so that the postsynaptic current is inward) or inhibitory (with V_{syn} below V_{rest} , say equal to V_K).

We begin with a simple case in which R_1 and R_2 both deliver brief excitatory input to the target neuron. The initiation times of $V_1^R(t)$ and $V_2^R(t)$ are respectively t_1 and t_2 . In our examples, we will fix t_2 and vary t_1 , where $t_1 = t_2 + \Delta t$ so that the input from R_1 precedes by $|\Delta t|$ the input from R_2 (if Δt is negative; a positive value of Δt means that R_2 fired before R_1). Suppose that the synapses are excitatory and identical, having the same synaptic weights and the same time courses except for initiation times. Assume that each input is subthreshold, but when delivered simultaneously they can fire the target cell. We see in Figure 6.9A that when R_1 fires far earlier than R_2 then two nearly identical input current events are seen in the target cell, and neither leads to a target cell firing. No nonlinearities are evident in this case. However, when the inputs are delivered within a restricted time window they can cause the target cell to fire (Figure 6.9B,C). Thus the neuron's integration and threshold properties make it act like a coincidence detector for excitatory input. The regenerative spike-generating

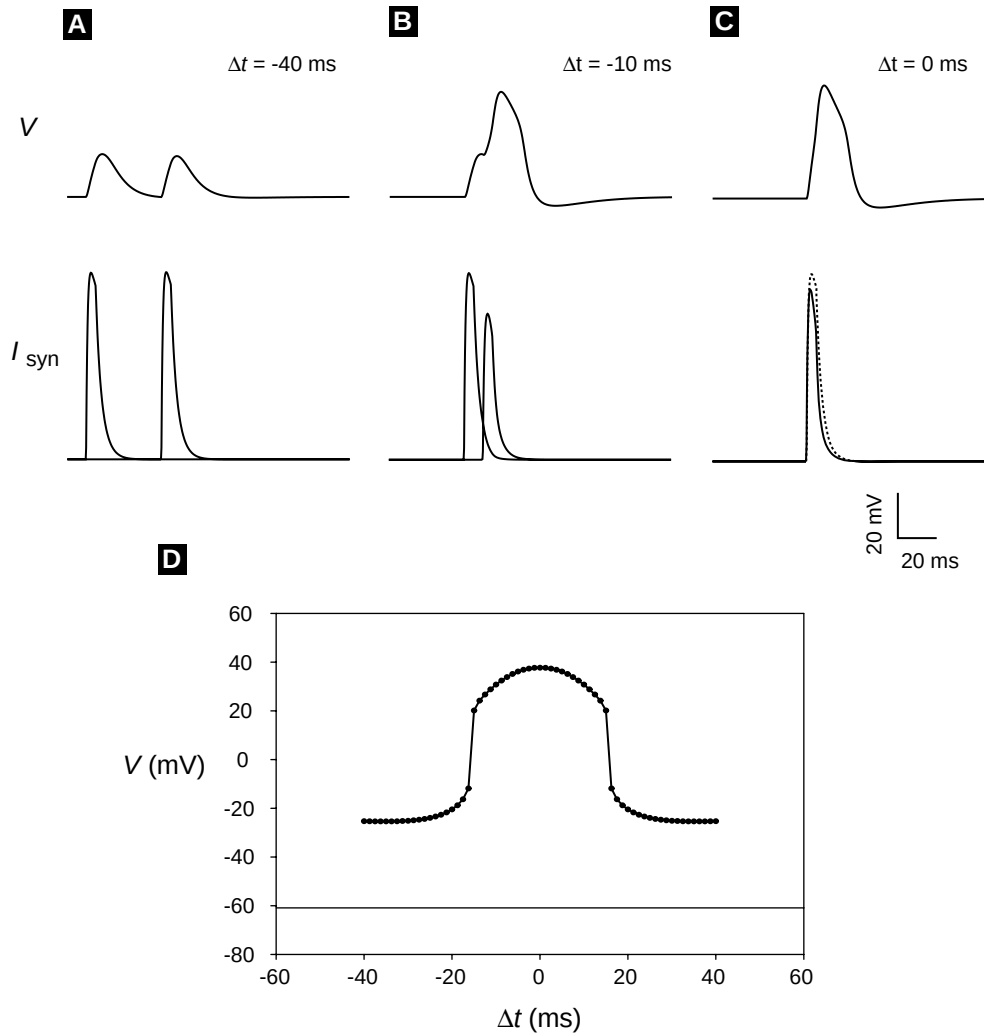


Figure 6.9 Coincidence detection for excitatory inputs. $V(t)$ time courses of target neuron (upper records) for excitatory inputs (lower records are synaptic currents) from receptor cells R_1, R_2 that are offset in time by Δt . Results for three different values of Δt (in ms): -40 in (A), -10 in (B), 0 in (C). Note: The current record shown dashed in (C) is reproduced from (A) to show the effect of reduced driving force when inputs occur simultaneously. (D) Response plot of peak of $V(t)$ (following second input) versus Δt . Parameter values for excitatory synaptic inputs: $\alpha_1 = \alpha_2 = 1.0/\text{ms}$, $\beta_1 = \beta_2 = 0.3/\text{ms}$, $V_{\text{syn},1} = V_{\text{syn},2} = 100$ mV, $\theta_{\text{syn}} = 20$ mV, $k_{\text{syn}} = 2$ mV; $g_1 = g_2 = 1.0$ mS/cm². At $t = 0$ neurons are at rest, $V = -60.9$ mV, $w = 0.0149$ and synaptic gating variables are zero.

mechanism is quite nonlinear; its output does not reflect linear summation of inputs: one plus one is not necessarily equal to two.

The response plot (Figure 6.9D) summarizes the sensitivity to Δt ; it is the tuning curve for the target cell as a coincidence detector. Notice how sharp the sensitivity is. If this target neuron activates synapses that feed higher order neurons and if the synapses require pre-synaptic depolarization above 0 mV, then the next level would effectively see the yes/no answer. Emphasizing the computational analogy one might say that the target neuron is performing the logical AND operation. Notice further that the tuning curve is symmetric about $\Delta t = 0$, as expected. Since the inputs are identical, the target cell's response is the same for R_1 preceding or following R_2 by Δt .

In Section 6.2 we demonstrated the feature of sublinear addition of synaptic inputs that are generated by conductance changes. The basis for this is that the synaptic current depends on the membrane potential through the driving potential $V(t) - V_{\text{syn}}$. Thus the current for a given postsynaptic conductance is reduced when $V(t)$ operates closer to V_{syn} . This effect is seen in Figure 6.9B for the case of nearly (but not exactly) simultaneous transient inputs. When $|\Delta t|$ is small enough the target neuron is still depolarized when the second input arrives. Hence current injected by this second input is smaller than that resulting from the earlier input. Similarly, when the two inputs occur simultaneously there is a reduction in the unitary input (compare solid with dashed records in the lower part of Figure 6.9C), but the effect is less pronounced, since the input current is complete before significant depolarization occurs.

Next we illustrate the vetoing power that inhibition can exercise over a suprathreshold excitatory input. For this case R_2 yields an excitatory conductance as in Figure 6.9 except doubled in size, adequate to evoke by itself a spike in the target neuron. Also, R_1 activates an inhibitory synaptic conductance that has a slower time course than an excitatory one. If inhibition is delivered sufficiently in advance of excitation, then it has no deleterious effect on the response to excitation (Figure 6.10A1). However, if it occurs in a narrow temporal window that starts just before excitation, it can overrule the excitation, vetoing an action potential in the target cell (Figure 6.10A2). Even if it occurs slightly after the excitatory input, it can still prevent spiking or at least shorten the target neuron's action potential (Figure 6.10A3). The tuning curve for this vetoing or subtraction effect is not symmetric (Figure 6.10A4). The inhibition with its slowly decaying tail is more effective if delivered with or reasonably before the excitation. But it is ineffective if delivered afterward even by a modest amount. Once the intrinsic spike-generating conductances are activated only a very strong and not too delayed inhibition can cancel the spike. Such an input-output relation might be likened to the logical AND-NOT operation: An output spike occurs only if R_2 fires and R_1 does not.

If one imagines that the receptor cells R_1 and R_2 are activated in temporal sequence (with delay Δt) by a moving stimulus in the environment, our target neuron shows sensitivity to the direction of this movement. Suppose that a stimulus moving from left to right activates R_1 and then R_2 . In this case our tuning curve shows that vetoing occurs far more effectively for movement from left to right (Δt negative) than from right to left (Δt positive). Now suppose that we want to go the next step by having an output neuron that acts as a directionally selective motion detector. Can we create a unit that fires only when the stimulus moves across the receptor pair from left to

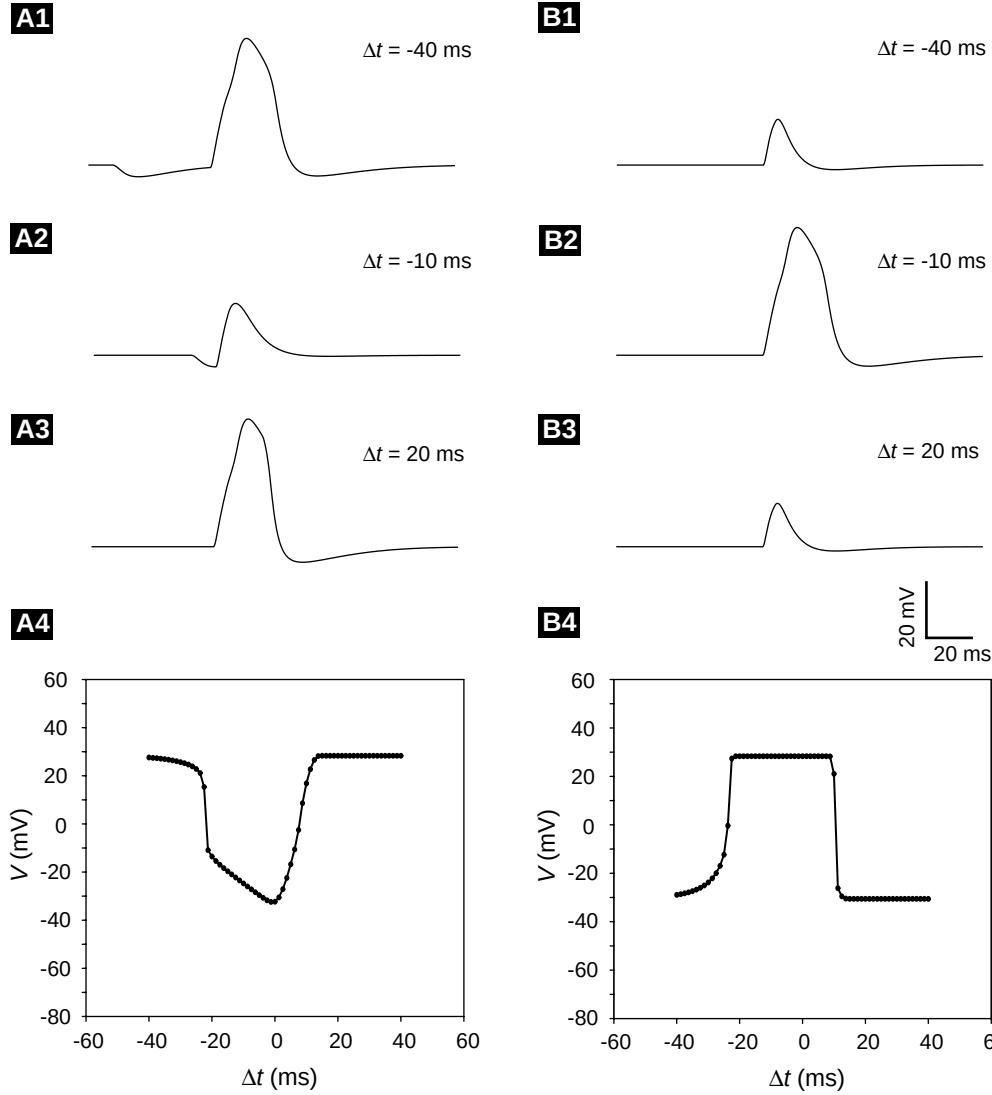


Figure 6.10 (A1–A4) Implementing subtraction with excitation and inhibition. (A1–A3) membrane potential time courses $V(t)$ of target neuron for excitatory input from receptor cell R_2 and inhibitory input from R_1 , offset in time by Δt (see schematic in Figure 6.8B). Results for different values of Δt , -40 , -10 , 20 ms, are shown in (A1–A3), respectively. (A4) Response plot of peak of $V(t)$ (associated with input from R_1) versus Δt . Parameter values for excitatory inputs as in Figure 6.9; for inhibitory input kinetics: $\alpha_1 = 0.5/\text{ms}$, $\beta_1 = 0.1/\text{ms}$, $V_{\text{syn},1} = -84$ mV, $\theta_{\text{syn}} = 20$ mV, $k_{\text{syn}} = 2$ mV, $g_1 = 2.5$ mS/cm², $g_2 = 2.0$ mS/cm². (B1–B4) Directionally selective, motion-detecting neuron. (B1–B3) membrane potential time courses $V_2(t)$ of second layer neuron (Figure 6.8C) neuron N_2 that receives excitatory input from receptor cell R_2 and inhibitory input from N_1 (A1–A4). Results for different values of Δt , -40 , -10 , 20 ms, are shown in panels (B1–B3), respectively. (B4) Response plot of peak of $V_2(t)$ versus Δt . Parameter values for N_1 and the inputs to N_1 are same as in the left panels; for N_2 and input kinetics and their weights onto N_2 the parameter values are identical as onto N_1 (with $g_{2,1} = 2.5$ mS/cm², $g_{2,R_2} = 2.0$ mS/cm². Input from R_2 to N_2 is delayed in each case by 10 ms. At $t = 0$ neurons are at rest, $V = -60.9$ mV, $w = 0.0149$ and synaptic gating variables are zero.

right and in a confined range of Δt (this means either that the stimulus is moving fast enough or it is assumed of moderate speed and in the spatially localized receptive fields of these receptors)? We want a tuning curve that looks like Figure 6.10B4.

A simple way to solve conceptually this design task is to imagine subtracting the tuning curve of Figure 6.10A4 from the response curve of R_2 delivered alone to the target neuron (the latter being just flat at 32 mV, independent of Δt). One way to implement this in neuralware is by introducing a downstream neuron N_2 (with voltage $V_2(t)$) whose tuning curve will have this property. Let us refer to the target neuron in earlier examples as N_1 (with voltage $V_1(t)$). The two Morris–Lecar units N_1, N_2 have identical intrinsic properties. To carry out the subtraction we now feed excitatory input from R_2 into N_2 , with some propagation delay, and have N_1 inhibit N_2 (Figure 6.8C). Thus for large $|\Delta t|$ the inhibition from N_1 vetoes the excitation from R_2 with the result that N_2 does not fire (Figure 6.10B1,B3). For modest $|\Delta t|$ the inhibition is preferentially absent for Δt negative and N_2 can fire (Figure 6.10B2). Figure 6.10B4 shows the tuning curve of the output neuron N_2 . With this configuration we identify N_1 and N_2 as neurons in the first and second processing layers after the network’s input layer of receptor units. Other types of feedforward connectivities can lead to different feature-detecting properties; e.g., in the primary visual cortex there are neurons (“simple cells”) that respond to bars of light with specific orientation, and the sensory signals coming to these neurons are mapped with specific alignments from the thalamus (which shows, to first approximation, no orientation specificity).

The examples above have exploited the sensitivity of the model neuron’s responsiveness to timing of its inputs. There is good evidence that sound localization as carried out by the auditory system (in some frequency ranges) can involve neurons capable of coincidence discriminations at submillisecond time scales. These neurons have been shown in some species to have potassium currents that are believed to render them unresponsive to tonic inputs but highly sensitive to transient inputs [Trussell, 1999]. Whether or not neural computations generally are performed based on precise spike timing raises interesting questions about the nature of the neural code and mechanisms [Shadlen and Newsome, 1998, Rieke et al., 1997, Dayan and Abbott, 2001]. How could spike timing be preserved in the face of neural variability, including the possible sources of heterogeneity of neuronal and coupling properties, sparseness of coupling, and the stochastic nature of synaptic transmission? Can some type of population averaging or multi-layer processing recover timing information, or must it necessarily be degraded as activity passes through successive layers? In some contexts firing rate (short time averaged) is likely the code carrier.

6.5 Large-Scale Networks

The previous sections have dealt with simple 2 or 3-cell networks, but how does one develop and implement formulations to treat large cellular ensembles? One approach, of course, is just to extend to N units the cell-based descriptions that we have been using

above. This means specifying the biophysical details of each unit, the coupling dynamics (e.g., synaptic conductances or diffusion of intercellular messengers), and the coupling topology (e.g., nearest-neighbor or all-to-all coupling or something intermediate). In the case of spiking units this would mean that we are attempting to keep track of each spike in the network and time course of intrinsic variables (membrane potential and gating variable for ionic conductances at all times) and coupling variables associated with the intercellular communication. Such approaches have been used to model electrical activity in the heart [Keener and Sneyd, 1998], the dynamics of hormonal secretory systems like the pancreas [Sherman, 1996], and behaviors of neural systems like epileptic waves in brain slices [Traub et al., 1999] or receptive field properties of visual cortex [Troyer et al., 1998, McLaughlin et al., 2000].

An alternative approach to such detailed cell-based models is to use mean field descriptions. For this one assumes that the space and time scales of interest do not demand knowing what each individual unit is doing at every moment in time; that the amount of heterogeneity and fluctuation in the system is enough but not too much so that averaging over cellular properties and small time and space scales is allowable to yield a description for the dynamics of the ensemble averages. For example, instead of instantaneous membrane potential (for each cell, on the time scale of milliseconds) being the primary observable, we might use the firing rate of a cell on the time scale of tens of milliseconds and a spatial scale that includes several cells in a unit length. If you think about it, the Hodgkin-Huxley model of electrical excitability is a mean field description at the cellular level with respect to the population of ionic channels. The random openings and closings of individual channels are not described; only the probability or fraction of gating variables being in a particular state is represented (see Chapter 11, as well as [Johnston and Wu, 1995] and [Koch, 1999]). An implicit assumption in using the mean field is that the individuals are behaving asynchronously. We would not use the HH model if we thought that the channels were all opening and closing precisely together in lock step. Similarly, using a mean field model for a cellular network means that we assume that the cells are not synchronized on a fine temporal scale, say spike for spike. Consequently, we would not use the mean field models to answer questions about precise spike timing, as we have been doing in the previous sections.

So how does one obtain the mean field model for an ensemble? This is a difficult and significant question, or several questions, actually: How do you retain some information about biophysical properties of cells and synapses when you go from a “micro” to a “macro” description? Can you be sure if you do a cell-based network simulation that your units will actually behave asynchronously? How do you decide whether your mean field model should retain explicitly some form of cell heterogeneity? How does one average over the different sources of stochasticity? These issues will keep researchers in computational cell biology quite busy over the next several years.

In practice, mean field or population-averaged models in cell/neuro biology are typically not derived from first principles. They are more or less developed from a set

of assumptions that involve disregarding some of the difficult issues. One may lose some connection with the biophysical details in this way.

Here's one example along these lines of a derivable mean field description. Suppose we treat a spike as an event without considering the ionic currents that generate the spike. The leaky integrate-and-fire model of neuronal excitability does this. Here one assumes that below “firing threshold” the neuron has a linear I-V relation and the membrane potential is described by an RC circuit:

$$C_m \frac{dV}{dt} = -\frac{V}{R_m} - I_{\text{syn}} + I_{\text{app}}, \quad (6.17)$$

where I_{syn} is the summed synaptic input from the other cells making synaptic contact with this one.

When V reaches threshold V_{th} then we say that a spike occurred, and V is reset to some level, typically to V_{rest} . In a case like this the model is linear except for the resetting events, and on the face of it each cell has one variable, V . So, what about the synaptic variables? One could assume a very simple model for a synaptic input that given a presynaptic spike the postsynaptic V jumps instantaneously up or down by a specified amount. This assumption throws away the time course of the postsynaptic conductance. However, on the plus side one may now derive legitimately a mean field population model, getting a (partial) differential equation for the probability density function $p(v, t)$ of membrane potential (say for $V_{\text{rest}} \leq V \leq V_{\text{th}}$) as a function of time. This description does indeed well describe a network of n such integrate-and-fire cells, sparsely coupled, in the limit of large n . It has been used to describe a variety of behaviors [Nykamp and Tranchina, 2000, Omurtag et al., 2000]. However, as one seeks to restore some of the biophysical variables, the description can become impractical. Each new dynamic variable, synaptic or intrinsic, that one includes becomes another independent variable in $p(v, \dots, t)$ leading to a high-dimensional system growing geometrically in size. In contrast, the problem size grows linearly if one adds additional variables in the case of an n -cell integrate-and-fire system. One of the insights that emerge from approaches like these is that if cells are really asynchronous, then at each instant in time they will have V -values spread over the interval, and one should expect that some fraction (perhaps small) of cells are near to V_{th} . Consequently, a transient excitatory input will lead to almost instantaneous firing of some cells. This means that the network can exhibit very fast response times for some inputs.

A less rigorously supported but attractive mean field model was popularized by Wilson and Cowan [Wilson and Cowan, 1972]. This formulation jumps right to mean field quantities, which in the simplest interpretation are the probabilities of firing (per unit time) of the typical, or “mean,” cell in the given populations: $E(t)$ and $I(t)$ for, say, excitatory and inhibitory subpopulations of interacting neurons. Note that this Wilson–Cowan formulation implicitly assumes random, sparse coupling without spatial structure. One assumes that the neurons have input/output relations represented by f_E and f_I so that under steady input to, say, the “ E ” cells, they would fire with steady rate $f_E(\text{input}_E)$ (and analogously for “ I ” cells). The shape of f_E is usually taken to be

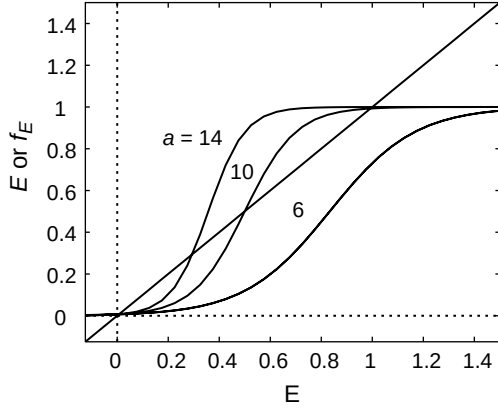


Figure 6.11 Graphical depiction for determination of steady states to single population of recurrently connected excitatory neurons. Left- and right-hand sides of (6.20), plotted as functions of E , firing rate. Definitions: $f_E(u) = 1/[1 + \exp(-u)]$ and $p_E = 5$; right-hand side plotted for three different values $a = 6, 10, 14$.

sigmoidal, the rate increasing to a saturation level as input E grows large. The network is considered not able to equilibrate instantaneously to dynamic inputs but that it “integrates” with a time constant τ_E .

For now, consider an isolated population of mutually excitatory cells. We would have

$$\tau_E \frac{dE}{dt} = -E + f_E(\text{input}_E), \quad (6.18)$$

where input_E includes input to the E population from both extrinsic sources P_E and from the other cells in the network. In the mean field approach the other cells are also firing at rate E , so the synaptic field is proportional to E . Thus, $\text{input}_E = a \cdot E + P_E$. Note that the parameter a incorporates the transformation from presynaptic firing rate to postsynaptic conductance and to postsynaptic current, etc. One should also appreciate that in using the sigmoidal-like input/output relation there is not a strict threshold for cell firing. The firing probability decreases toward zero for decreasing input strength, but it is never really zero even for negative inputs. An implicit assumption here is that noise in the system smoothes the abrupt thresholds as found in frequency-current relations of, say, cellular models like Morris-Lecar without noise. Also, in any real network cells are not perfectly identical, so this heterogeneity also contributes to such smoothing in the mean field models.

Now let us warm up by considering just the isolated E network and ask what steady states of activity are possible. At steady state we must have

$$\frac{dE}{dt} = 0, \quad (6.19)$$

so that from (6.18),

$$E = f_E(a \cdot E + P_E). \quad (6.20)$$

The solutions to this nonlinear transcendental equation are most easily analyzed graphically. That is, plot the left-hand side and the right-hand side on the same axes as

functions of E . Their intersections correspond to the possible solutions. The parameter a acts primarily on the steepness of the sigmoidal curve, while P_E is more related to left/right translation. In Figure 6.11 we can see that for each value of a there is an intersection near $E = 0$. That is, the network has a stable rest state for these parameter values. On the other hand, we see that if a is large enough, then there are multiple steady states. For this one-dimensional dynamical system the middle state of moderate activity is unstable, while the upper state of high activity is stable. In this parameter regime the network is bistable. If the cells are at rest (very low firing rate) and no transient input is given, then they will remain at rest. However, if a sufficiently strong brief input is delivered, then the cells will become active and because of the recurrent excitation in the network the cells will continue to drive each other, and the network reaches a steady state of high activity. This recurrent excitation is the autocatalytic mechanism that underlies the bistability. We may view a as the relative weight of the autocatalytic feedback, the strength of excitatory interactions. Note that because the input/output relation is saturating the activity level does not grow without bound.

Now we are ready to consider the two populations of excitatory and inhibitory cells. Extending the development above, we write down the model equations as

$$\tau_E \frac{dE}{dt} = -E + f_E(a \cdot E - b \cdot I + P_E), \quad (6.21)$$

$$\tau_I \frac{dI}{dt} = -I + f_I(c \cdot E - d \cdot I + P_I). \quad (6.22)$$

This pair of ordinary differential equations governs the behavior, in the mean field sense, of the interacting E and I populations. Note the signs of the interaction terms inside of f_E and f_I . Positive a means recurrent excitation, autocatalysis; positive c means that I is driven by E ; positive $c \cdot b$ (with the minus sign on b) means that negative feedback through I onto E ; and $-d \cdot I$ means I cells inhibit each other.

The model equations can be analyzed with phase plane methods. For the case shown in Figure 6.12 the network has a stable rest state (the only intersection of the E - and I -

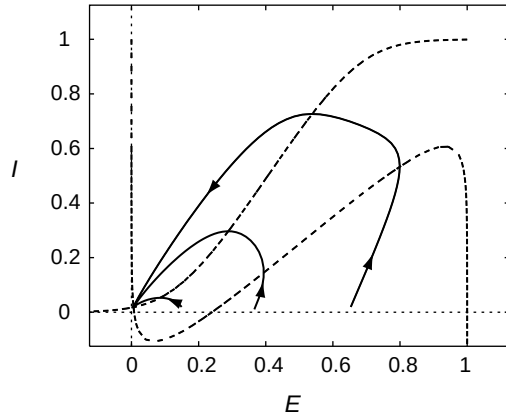


Figure 6.12 Network excitability. Phase plane portrait for firing-rate model of neuronal network with interacting excitatory and inhibitory subpopulations, based on (6.22). The cellular input/output relations (rate vs. input) are $f_E(u) = f_I(u) = 1/[1 + \exp(-u)]$ as in Figure 6.11. The E -nullcline is the cubic-shaped dashed line; I -nullcline is the sigmoidal dashed line. Trajectories (solid), for three different initial conditions that represent perturbations from rest to the E cells, all tend back to the rest state for large time. Parameter values: $a = 16, b = 12, c = 16, d = 5, P_E = -5, P_I = -4, \tau_E = \tau_I = 1$. Note that the model equations have been expressed in dimensionless form.

nullclines) with both populations at very low firing rate. This state is globally attracting. For any initial conditions the network has a transient response and then returns to rest. The three different trajectories correspond to responses of E and I for brief stimuli to the E population. For a small stimulus only a few E cells fire, not enough to amplify the excitatory activity. However, with a larger initial level of E we see some regenerative effects. First, E grows beyond its starting value: Some nonlinear amplification occurs. Then, E peaks as the I population is recruited, driven by the firing of the excitatory cells. Eventually, all cells return to the rest state. This illustrates the excitability of the network.

The phase plane portrait of the network is qualitatively similar to that of the basic Morris–Lecar model of cellular electrical excitability. The activator variable (in this case E) has a cubic-shaped nullcline; the left, middle, and right branches correspond to the three roots of (6.20) for a fixed level of $P_E - b \cdot I$. The variable I here corresponds to a recovery process like w in the Morris–Lecar model. Its nullcline is monotonic.

If we vary the parameter P_E , the E -nullcline changes, while the I -nullcline remains fixed. For example, increasing a bit the extrinsic input to the E population, say imagining more feedforward excitation onto the E cells from some other network, will cause them to fire faster, and thereby recruit more inhibition. The steady state will migrate to the right and upward along the I -nullcline. If one increases P_E enough, then the steady state will move onto the middle branch and perhaps destabilize. This is seen in Figure 6.13A, where the network is now in an oscillatory mode, having gone through a Hopf bifurcation. The negative-feedback inhibition lags somewhat behind the excitatory activity, seen in this phase plane portrait as well as in the time courses shown in Figure 6.13B. If the E -population is driven hard enough, the system stops oscillating. The steady state, for large enough P_E , restabilizes and is on the right branch of the E -nullcline. This corresponds to a balanced state with strong steady firing in each population.

These examples are only illustrative of this class of mean field firing-rate models. Some specific neural systems where such formulations have been applied include [Pinto et al., 1996, Ermentrout, 1998, Hansel and Sompolinsky, 1998]; in the paper by Pinto et al. you will find a pseudo-derivation of the model based on cellular properties.

Suggestions for Further Reading

- *Spikes*, Fred Rieke, David Warland, Rob de Ruyter van Steveninck, and William Bialek. This influential book treats neuronal communication from a highly information-theoretic point of view [Rieke et al., 1997].
- *Neural networks as spatio-temporal pattern-forming systems*, Bard Ermentrout. This paper reviews pattern formation in neural networks, including the continuous networks discussed in this chapter [Ermentrout, 1998].
- *Mathematical Physiology*, James Keener and James Sneyd. Keener and Sneyd treat some of the cellular level topics presented in this chapter, as well as many other

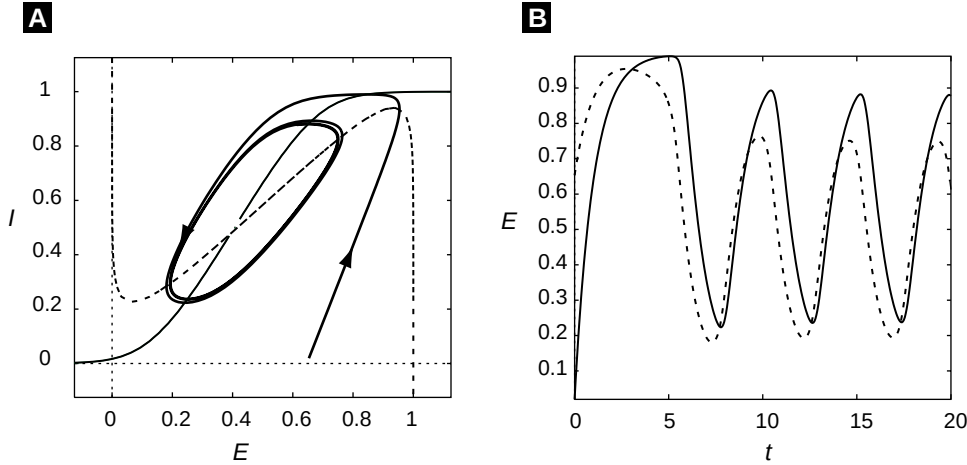


Figure 6.13 Network oscillations. (A) Phase plane and (B) time courses for the firing rate model, with increased input to the excitatory population: $P_E = -1$. Other parameters are as in Figure 6.12. Dashed line is for E -nullcline in (A) and E vs. t in (B); solid line is for I -nullcline in (A) and for I vs. t in (B). Darker solid curve in (A) is the phase plot.

topics in systems level physiology, from a more analytic perspective as opposed to the computational focus presented here [Keener and Sneyd, 1998].

- *Methods in Neuronal Modeling*, Christof Koch and Idan Segev, editors. This is a compilation of chapters from various authors on a wide variety of topics related to neuronal modeling, both at the cell and at the network level [Koch and Segev, 1998].
- *Foundations of Cellular Neurophysiology*, Daniel Johnston and Samuel Wu. Covers in more depth topics such as transmitter release, plasticity, elementary networks and extracellular electrical behavior [Johnston and Wu, 1995].
- *Theoretical Neuroscience: Computational and Mathematical Modeling of Neural Systems*, Peter Dayan and Larry Abbott. This book addresses neural coding issues with several different types of models [Dayan and Abbott, 2001].
- *Biophysics of Computation*, Christof Koch. This book covers many aspects of cell-based neurophysiology from a modeling perspective [Koch, 1999].

Exercises

1. Consider the dynamics of two identical electrically passive cells with gap-junctional coupling:

$$\frac{dV_1}{dt} = (-g_m * V_1 + g_c * (V_2 - V_1) + I_1) / C_m,$$

$$\frac{dV_2}{dt} = (-g_m * V_2 + g_c * (V_1 - V_2) + I_2)/C_m.$$

Let $g_m = 0.05 \text{ mS/cm}^2$, $C_m = 1 \mu \text{ F/cm}^2$. Consider $g_c = 0.1 \text{ mS/cm}^2$ (Note: We have divided the equations by cell area). For now, let $I_1 = I_2 = 0$.

- (a) Solve the equations with $V_1(0) = 1, V_2(0) = 0$ and observe, over 60 ms, the convergence and eventual decay back to rest of both cells.
 - (b) This is a linear system. Its solution is a sum of two exponentials. Find the characteristic decay rates (eigenvalues) and the corresponding time constants. (Hint: Find singular points.)
 - (c) Evaluate and plot $V_s = (V_1 + V_2)/2$ and $V_d = (V_1 - V_2)$. Notice that they appear to be single exponentials. Confirm this by plotting V_s and V_d on a semilog scale and identify the slopes with the eigenvalues.
 - (d) Derive differential equations for V_s and V_d by adding and subtracting the equations for V_1 and V_2 . See that these equations are uncoupled, and so each of V_s and V_d decays as a single exponential. Identify the eigenvalue for V_d as the reciprocal of the system's equalization time constant.
 - (e) Do the eigenvalues change if I_1 and/or I_2 are nonzero?
2. Determine the two time constants of the following system: See that compartment 1 acts as a driver for compartment 2:

$$\begin{aligned} \frac{dV_1}{dt} &= (-g_m * V_1 + \alpha * g_c * (V_2 - V_1) + I_1)/C_m, \\ \frac{dV_2}{dt} &= (-g_m * V_2 + g_c * (V_1 - V_2) + I_2)/C_m. \end{aligned}$$

Let $g_m = 0.05 \text{ mS/cm}^2$, $C_m = 1 \mu \text{ F/cm}^2$, $g_c = 0.1 \text{ mS/cm}^2$. Noting that we have divided the equations by cell area, α is the ratio of areas: Cell 2's area divided by cell 1's. Consider $\alpha = 1$ or 0.1 . For now, let $I_1 = I_2 = 0$. Note: In this formulation g_c is the total gap junctional conductance divided by the area of cell 2.

- (a) Compute and plot the time courses for V_1, V_2 (on the same axes) for the two values of α . Interpret the results. Why does V_1 decay more slowly for $\alpha = 0.1$? What happened to the early decaying component?
- (b) Estimate (analytically) the eigenvalues in the limit of α very small. Explain the differences between these eigenvalues for the two cases of α , 1 or small.
- (c) How well is the behavior of V_1 and V_2 approximated for small α if you just set $\alpha = 0$ in the equations? Discuss the behavior in terms of cell 1 being a driver of cell 2 (think of the larger membrane area in the dendrites). Now do the simulation with initial conditions $V_1(0) = 0, V_2(0) = 0$. How effectively does compartment 2 drive compartment 1?
- (d) Suppose the steady input I_1 is delivered to compartment 1. Find the attenuation factor for the steady response from the dendrite to the soma; V_1/V_2 . Do the

same for I_2 given to compartment 2. Compare and discuss the two attenuation factors and their differences.

3. Consider two Morris-Lecar cells with gap-junction coupling; use the same intrinsic params as in Figure 6.2 and Figure 6.3 (except $I_1 = I_2 = -5$, unless stated otherwise).
 - (a) Formulate the equations, as in Exercise 2, allowing for possibly different areas.
 - (b) Consider first an isolated cell. Show that it is bistable, with an attracting oscillation mode in the depolarized regime and a stable steady state at low membrane potential (look at and describe the phase plane: nullclines, steady states, and their stability, etc.). Demonstrate this bistability over a range of I values.
 - (c) Show that for weak gap-junction conductance g_c , the pair can exist with one cell, say cell 1, in the “upper” (oscillating) state and cell 2 in the lower (steady) state. But as g_c increases to intermediate values (say beyond $g_c = 0.3$), the cells tend to be in the same state. If g_c is not too large (say < 0.3 but > 0.1) the cells can oscillate in antiphase but for large g_c the cells oscillate in-phase. Show that there is some range of g_c where both oscillating states coexist.
 - (d) Consider the effect of unequal areas. Suppose cell 1 (as in Exercise 2) is larger and aratio is the ratio of their areas (as defined in Exercise 2): $\text{aratio} = \text{area}_2 / \text{area}_1$. Let $g_c = 0.3$. Verify that the state of cell 1 up (and oscillating) with cell 2 down is still valid. Compare its frequency to that of the control case ($\text{area ratio} = 1$) and explain the difference. Of course can also get the state w/ both up and oscillating. But now with a different phase relationship than control. In what way and why? Is there an analog here for the two states of in-phase and anti-phase as in (c)? Give an example if you say Yes.
4. We have discussed the Wilson-Cowan paradigm in excitable and in oscillatory modes.
 - (a) Using a numerical package, compute the bifurcation diagram with stimulation to “E” cells as a parameter. I think you be SNIC at one end and Hopf at the other... have students do this and
 - (b) Plot frequency vs stimulation and examine the low frequency behavior.
 - (c) Can you find other parameters that would give a bistable network?
5. Consider just an excitatory population. As a function of input, this population shows a range of bistability.
 - (a) Introduce a slow negative feedback, Q , as an intrinsic property. Q should grow with the firing rate of the excitatory cells, and result in spike frequency adaptation. The Q dynamics could be something like

$$\frac{dQ}{dt} = (Q_\infty(E) - Q) / \tau_q$$

The sigmoid Q_∞ should be positioned in the E-Q plane to create oscillations or excitability. This is one way to give the single population some interesting dynamics in the context of a rate model like Wilson-Cowan.

- (b) Try using θ_Q (the location of $Q_\infty = 0.5$) as the slow variable.